Shehtab Zaman

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Education

Binghamton University, State University of New York

Ph.D in Computer Science — Expected 2025

Master of Science in Computer Science - 3.9/4.0 — 2019 - 2020

Bachelor of Science in Mathematical Physics with a Minor in Computer Science - 3.9/4.0 — 2015 - 2019

Skills

Proficient Python, C++, CUDA, NCCL, NVSHMEM, PyTorch, Scikit-learn, Huggingface, MPI

Professional Experience

CS Department - Binghamton University - *Graduate Research Assistant* **December 2019 - Present**

- Led the development and evaluation of deep learning models (including physics-informed GNNs, LLMs, and Transformers) for critical molecular property prediction, synthesis, and retrosynthesis applications
- Pre-trained and fine-tuned transformers, GNNs, and generative models for inverse design and generation of molecules
- Creator of DGraph, an open-source large-scale distributed GNN training framework, capable of scaling beyond 100s of GPUs and learning on graph with a billion vertices
- Designed and currently maintain an open-source library for 3D grid generation, facilitating advanced molecular property prediction research
- Led the development of generative models (3D GANs and VAEs) for the inverse design of novel Metal-Organic Frameworks (MOFs), accelerating materials innovation
- Engineered deep neural network architectures, including 3D-CNNs and GCNNs, to enable high-throughput property characterization of Metal-Organic Frameworks (MOFs)

CS Department - Binghamton University - *Co-Instructor - Deep Learning* **August 2022 - Present**

- Created and delivered weekly lectures on fundamentals of deep learning (DL) for the graduate level Intro to DL course
- Designed assignments on optimization algorithms, neural networks, computer vision, and generative models

Lawrence Livermore National Laboratory - *Computing Scholar* **May 2022 - August 2022**

- Parallelized graph neural network via tensor parallelism to support large-scale graphs on the Livermore Big Artificial Neural Network (LBANN) toolkit
- Implemented GPU kernels with GPU-initiated partitioned global memory accesses for Gather-Scatter operations using NVidia's NVSHMEM library for distributed parallelized graph neural networks
- Performed kernel profiling and tooling on multi-GPU distributed CUDA kernels using NSight Systems and Compute

Lawrence Livermore National Laboratory - *Computing Scholar* **May 2021 - August 2021**

- Developed and implemented custom CUDA kernels for the large-scale deep learning training toolkit, enabling support for Graph Neural Networks and significantly improving computational efficiency.
- Optimized data pipelines and CUDA kernels to achieve scalable training of Graph Neural Networks on up to 100 GPUs, resulting in a 12x and 93x over speedup compared to PyTorch on 16 GPUs and 100 GPUs and enabling the prediction of DFT energies for 3 million samples in under 30 seconds.

Lawrence Livermore National Laboratory - *Data Science Institute - Research Scholar* **May 2020 - August 2020**

- Investigated scalable geometric and 3D deep learning with the LBANN toolkit on GPU accelerated HPC systems
- Implemented distributed, GPU accelerated Graph Convolutional Neural Networks in Python and C++
- Designed custom data ingestion pipelines for Numpy and HDF5 files for data-intensive workloads on HPC systems

Honors & Awards

Robert J Penfield Award - *Undergraduate Award for Junior Physics Student* **April 2018**

- Awarded due to outstanding academic promise in physics

Publications & Posters

BOOM: Benchmarking Out-of-distribution Molecular Property Predictions of Machine Learning Models — **May 2025**

Evan R. Antoniuk, Shehtab Zaman*, Tal Ben-Nun, Peggy Li, James Diffenderfer, Busra Demirci, Obadiah Smolenski, Tim Hsu, Anna M. Hiszpanski, Kenneth Chiu, Bhavya Kallkhura, Brian Van Essen*
Under Review, 2025

Consistent Idempotent Generative Models — **May 2025**

Shehtab Zaman, Patrick Chen, Xingye Qiao, Kenneth Chiu
Under Review, 2025

PtychoFormer: A Transformer-based Model for Ptychographic Phase Retrieval — **October 2024**

Ryuma Nakahata, Shehtab Zaman, Mingyuan Zhang, Fake Lu, Kenneth Chiu
Under Review, 2024

STRIDE: Structure-guided Generation for Inverse Design of Molecules — **October 2023**

Shehtab Zaman, Denis Akhiyarov, Mauricio Araya-Polo, Kenneth Chiu
AI for Scientific Discovery Workshop at NeurIPS 2023

ParticleGrid: Enabling Deep Learning using 3D Representation of Materials — **October 2022**

Ethan Ferguson, Shehtab Zaman*, Cecile Pereira, Denis Akhiyarov, Mauricio Araya-Polo, Kenneth Chiu*
IEEE 18th International Conference on eScience (eScience) 2022

Parallelizing Graph Neural Networks via Matrix Compaction for Edge-Conditioned Networks — **May 2022**

Shehtab Zaman, Tim Moon, Tom Benson, Sam Adé Jacobs, Kenneth Chiu, Brian Van Essen
22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid) 2022

MoleculeFlow: A Deep Generative Workflow for 3D Molecular Generation — **December 2021**

Ethan Ferguson, Shehtab Zaman*, Mauricio Araya-Polo, Denis Akhiyarov, Kenneth Chiu, Cécile Pereira*
ELLIS Machine Learning for Molecule Discovery Workshop

Graph Neural Network for Metal Organic Framework Potential Energy Approximation — **December 2020**

Shehtab Zaman, Christopher Owen, Kenneth Chiu, Michael Lawler
ML for Molecules Workshop at NeurIPS 2020

Real-Space Visualization of Quantum Phase Transition by Network Topology — **July 2019**

Shehtab Zaman, Wei Cheng Lee
Physical Review E Vol. 100 Issue 1

Towards Run Time Estimation of the Gaussian Chemistry Code for SEAGrid Science Gateway — **July 2019**

Angel Beltre, Shehtab Zaman, Kenneth Chiu, Sudhakar Pamidighantam, Xingye Qiao, Madhusudhan Govindaraju
Practice & Experience in Advanced Research Computing Conference

Presentations

ParticleGrid: A Library for 3D Molecular Representation for Deep Learning — **March 2022**

Shehtab Zaman, Ethan Ferguson, Kenneth Chiu, Mauricio Araya, Denis Akhiyarov, Cecille Pereira
APS March Meeting 2022

Graph Neural Network for Metal Organic Framework Potential Energy Approximation: Energy Landscape Database and Rigidity — **March 2021**

Christopher Owen, Shehtab Zaman
APS March Meeting 2021

Towards Inverse Design of Metal-Organic Frameworks to Maximize Hydrogen Storage using Deep Learning — **March 2021**

Kevin Phillips, Shehtab Zaman, Kenneth Chiu, Michael Lawler
APS March Meeting 2021